iFiNN

Intelligent Financial Neural Network

Democratizing AI-Powered Trading Intelligence

- 🎛️ No-Code AI Model Tuning
- 📊 Social Sentiment Analysis
- 🏆 Open Model Leaderboard
- 🛒 Strategy Marketplace

RECOGNIZED BY

Team Synapse · IISER Kolkata

The Gap in the Market

- ✗ **Black-Box AI Tools**
  Retail traders get predictions without understanding or control
- ✗ **No Hyperparameter Access**
  Users cannot tune models to their strategy or risk profile
- ✗ **Fragmented Tooling**
  5+ platforms needed: charts, news, ML, backtesting
- ✗ **Institutional Pricing**
  Bloomberg: ₹20L+/year — Inaccessible to retail
- ✗ **Information Overload**
  Thousands of news sources; no semantic understanding

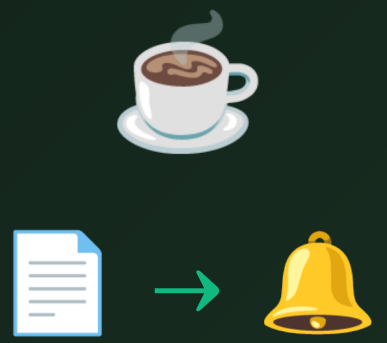
Our Solution

- ✓ **Transparent ML Pipeline**
  See exactly how predictions are made
- ✓ **Slider-Based AI Tuning**
  Adjust hyperparameters visually — no coding
- ✓ **All-in-One Platform**
  Charts + News + ML + Backtest + Execution
- ✓ **Accessible Pricing**
  Starting Free; Pro at ₹499/month
- ✓ **Semantic Analysis Engine**
  Context-aware news from Reddit, Twitter, Discord

Picture this:
We've Built It Before You Read About It

You're sipping your morning coffee, reading about a groundbreaking new equation in a research paper on quantitative finance.

Before you finish reading, iFiNN sends you a notification:
"This algorithm is now live on your dashboard."



🔔 New Feature Alert

"Fama-French 5-Factor Model now available in your ML pipeline"

⚡ Research-Driven Development

We track arXiv, SSRN, and quantitative finance journals daily. New papers = New features.

🧪 Bleeding-Edge Algorithms

From academic theory to production code in weeks, not years. Stay ahead of the market.

🎯 Zero Learning Curve

Complex models made simple. If it's published, we've packaged it for one-click deployment.

No-Code AI Model Tuning Interface

Anyone can train professional-grade ML models using simple sliders — no Python, no code, no complexity.

Visual Hyperparameter Control

Learning Rate 0.001

Hidden Layers 3

Dropout Rate 0.2

Epochs 100

Available Models

LSTM Networks

Long Short-Term Memory for sequential price patterns

GRU Networks

Gated Recurrent Units for faster training cycles

Transformer Encoder

Attention-based architecture for complex patterns

XGBoost / LightGBM

Gradient boosting for feature-rich predictions

Why It Matters

Traditional platforms give you a number. We give you the **power to build your own intelligence** — tuned to your strategy, your risk tolerance, your market view.

📱 Real-Time Social Sentiment Intelligence

Semantic analysis of Reddit, Twitter/X, Discord, and Telegram — not keywords, but context.

Data Pipeline

1. Multi-Platform Ingestion
r/wallstreetbets, r/CryptoCurrency, FinTwit, Discord Alpha groups

2. NLP Processing (Transformer-based)
FinBERT sentiment scoring, entity extraction, topic clustering

3. Signal Generation
Aggregate trends, detect anomalies, identify emerging narratives

4. Price Correlation
Link sentiment shifts to historical price movements

Why Social First?

⚡ **Faster Than News**
Social discourse precedes mainstream coverage by hours

👥 **Crowd Wisdom**
Millions of retail opinions aggregated in real-time

🎯 **Trend Prediction**
Catch meme stocks, FOMO cycles, pump patterns early

Use Cases

Crypto: Detect CT sentiment shifts before pumps
Stocks: WSB monitoring for retail momentum
Macro: Fed commentary, geopolitical event detection

🎯 Custom Domain & Niche Market Analysis

Go beyond generic stock picks. Analyze any domain, any niche, any market segment.

🏥

Healthcare & Pharma

Track FDA approvals, clinical trials, drug launches. Sentiment on biotech subreddits.

🌱

Green Energy & ESG

Monitor renewable energy discourse, carbon credit trends, policy impacts.

🎮

Gaming & Metaverse

NFT sentiment, gaming token trends, virtual real estate market analysis.

🇮🇳

Indian Markets (NSE/BSE)

Native integration with Indian brokers. Option chain analysis, FII/DII tracking.

🌐

DeFi Protocols

TVL tracking, yield farming opportunities, protocol governance votes.

🏠

Real Estate REITs

Property market sentiment, REIT performance tracking, regional analysis.

Configurability

Users can define **custom keyword filters**, **subreddit watchlists**, and **Twitter handles** to create hyper-focused analysis pipelines for any niche they want to track.

System Stack

Frontend (React + Lightweight Charts)
Interactive dashboards, real-time streaming, responsive UI

API Gateway (FastAPI)
REST + WebSocket, Pydantic validation, rate limiting

Core Analytics (Python)
130+ indicators via pandas_ta, Fourier analysis, pattern detection

ML Engine (GPU-Accelerated)
TensorFlow, PyTorch, XGBoost with CUDA support

Data Layer
PostgreSQL + TimescaleDB, Redis caching, S3 storage

Platform Capabilities

130+ TECHNICAL INDICATORS	**18+** PATTERN DETECTORS
10+ ML MODELS	**<1ms** DATA LATENCY

Data Integrations

Crypto: Binance, Bybit, Coinbase, OKX
Equities: NSE/BSE (India), Alpha Vantage, Polygon
Alt Data: Reddit, Twitter/X, Discord APIs

🏆 Open Model Leaderboard

A public competition where users submit their ML models — the best performers rise to the top.

Rank	Model Name	Sharpe	MDD
🥇	AlphaNet-v3	2.84	-8.2%
🥈	MomentumLSTM	2.61	-11.4%
🥉	SentimentGBM	2.45	-9.7%
4	VolatilityNet	2.31	-14.2%
5	TrendFormer	2.19	-12.8%

Like Kaggle for Finance

Transparent benchmarking across 50+ assets. Models ranked by Sharpe, Sortino, MDD, Win Rate, and more.

How It Works

1. **Train Your Model**
   Use our platform or upload custom models
2. **Submit to Leaderboard**
   Automatic backtesting on held-out data
3. **Compete & Climb**
   Rankings update weekly based on live performance
4. **Monetize Your Model**
   Top models can be sold in the marketplace

🏅 Recognition & Rewards

Top creators get featured, earn badges, and gain followers. Build your reputation as a quant.

🛒 Strategy Marketplace

Buy, sell, and license trading models. The App Store for financial algorithms.

Marketplace Revenue Model

💰 **Model Licensing Fees**
Creators sell models; iFiNN takes 15-30% commission

📈 **Subscription Models**
Pay monthly for access to top-performing strategies

🔄 **One-Click Cloning**
Deploy any strategy to your account instantly

✅ **Verified Badges**
Certified backtested performance guarantees

For Creators

Build your model → Submit to leaderboard → Rank high → List on marketplace → Earn passive income

💎 Data is the New Gold

Every model trained on our platform generates valuable data: feature importance, prediction accuracy, market correlations.

We sit on the most valuable asset of the 21st century — structured financial intelligence.

Enterprise Licensing

Hedge Funds: Pay for exclusive access to top models
Banks: White-label the platform for their clients
Fintechs: API access to our sentiment data streams

Capability	iFiNN	TradingView	QuantConnect	Bloomberg
User-Configurable ML Models	√ **Full Control**	✗ **None**	Limited	✗ **None**
Social Sentiment Analysis	√ **Reddit/X/Discord**	✗ **None**	✗ **None**	Basic
Open Model Leaderboard	√ **Public**	✗ **None**	Private	✗ **None**
Strategy Marketplace	√ **Buy/Sell**	✗ **None**	Limited	✗ **None**
Indian Market Focus (NSE/BSE)	√ **Native**	Basic	✗ **None**	√ **Yes**
Annual Cost	**₹0 - ₹24K**	₹15K - ₹120K	₹0 - ₹10L+	**₹20L+**

Our Unfair Advantage

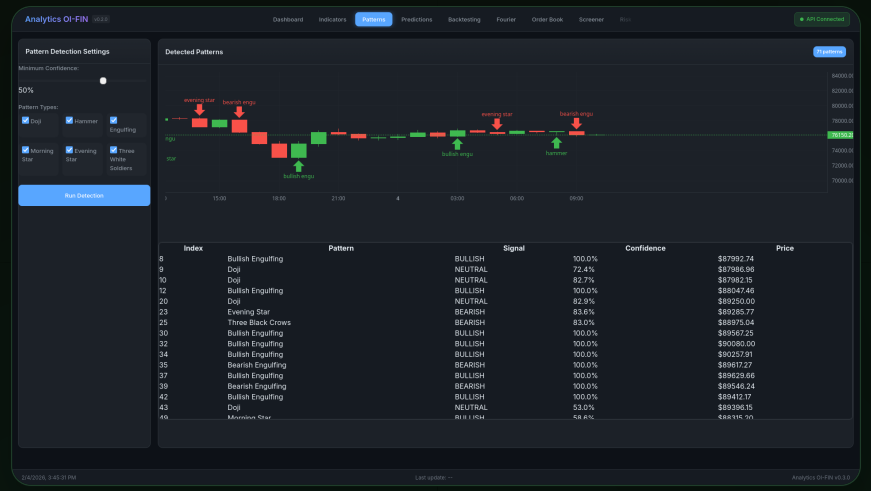
Transparency
Full ML pipeline control

Social-First
Reddit/X sentiment engine

Community
Leaderboard + Marketplace

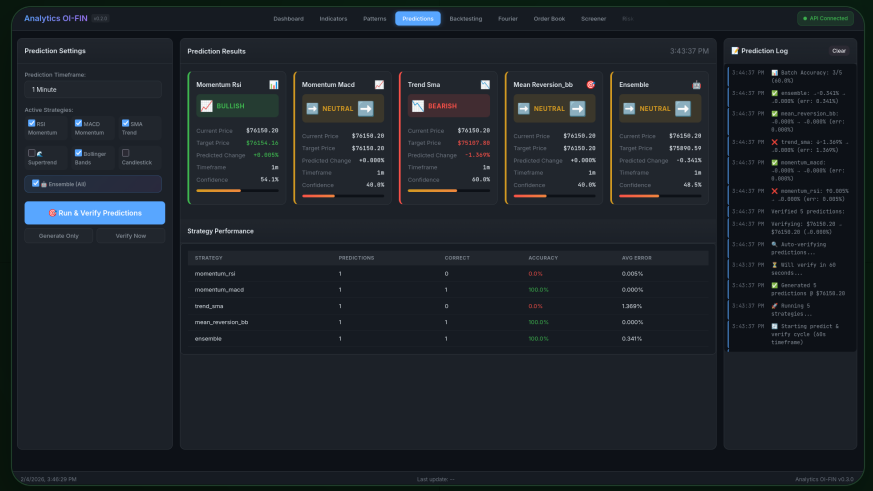
India Focus
Native NSE/BSE integration

Pattern Detection Engine



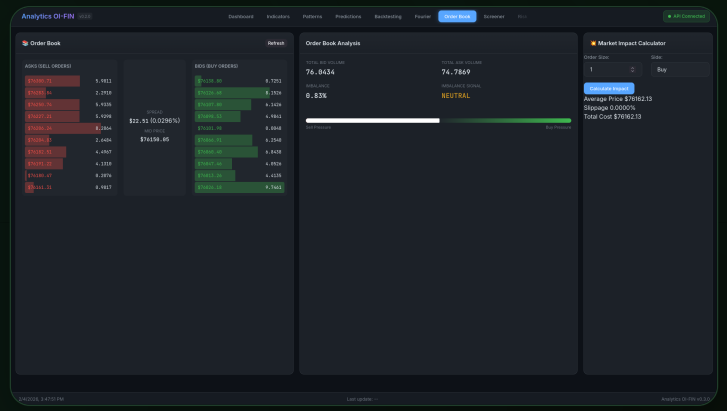
Automated candlestick pattern recognition

Live Prediction Dashboard



Real-time ML model performance tracking

Order Book Analytics (Work in Progress)



Live order flow and depth visualization

Societal Impact

Financial Democratization

Giving retail traders the same tools as hedge funds. Leveling the playing field.

Education & Transparency

Every model comes with theory documentation. Learn while you trade.

Reduce Retail Losses

Target: 30% improvement in retail trade success rates by eliminating emotional trading.

Market Size

100M+
INDIAN DEMAT ACCOUNTS

$150B
INDIAN FINTECH MARKET

18.2%
CAGR (2024-2030)

90%+
RETAIL LOSS RATE (SEBI)

Why Now?

Post-COVID trading boom · Record Demat account openings · Growing AI awareness · Crypto adoption in India · Need for transparent tools

Free Tier

₹0

- 30+ core indicators
- Basic pattern detection
- Paper trading
- Community access

Pro

₹499/mo

- 130+ indicators
- Real-time streaming
- Pre-trained ML models
- Unlimited backtests

Elite

₹1,999/mo

- Custom ML training
- GPU acceleration
- Sentiment API access
- Priority support

Enterprise

Custom

- White-label solutions
- On-premise deploy
- Dedicated support
- Custom development

Additional Revenue Streams

Marketplace Commissions:15-30% on model sales

API Credits:₹0.10 - ₹1 per call

Enterprise Data Licensing: Custom pricing

Educational Courses:₹999 - ₹4,999

Data is the New Oil

Every user interaction trains our models. Every prediction creates data. We're not just a platform — we're building the largest financial intelligence dataset in India.

Q2 2026

Foundation

- √ Core analytics engine
- √ 130+ indicators
- √ Pattern detection
- √ Backtesting engine
- ○ Frontend UI (80%)

Q3 2026

ML Integration

- ○ Hyperparameter UI
- ○ Model training pipeline
- ○ Sentiment engine
- ○ Social data feeds
- ○ Model explainability

Q4 2026

Beta Launch

- ○ Public beta (Free + Pro)
- ○ Leaderboard launch
- ○ Community features
- ○ Indian broker integration
- ○ Mobile app (beta)

2027

Scale

- ○ Marketplace launch
- ○ Enterprise tier
- ○ Live trading bots
- ○ Global expansion
- ○ API monetization

Current Status

Core analytics engine complete · FastAPI backend 90% done · Frontend UI in active development · Alpha testing with IISER Kolkata students

Join the Revolution

The future of financial intelligence is transparent, accessible, and community-driven.

synapse-iiserk.github.io

🌐 Live Website

📧 team@ifinn.tech

🐙 github.com/Synapse-iiserk

🎛️ No-Code AI

📊 Social Sentiment

🏆 Open Leaderboard

💎 Data = Gold

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